

SIGNA: Semantic Intent Guarantee and Notary Architecture — A Certification Framework for AI Output Trustworthiness

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Abstract

Current AI systems generate outputs at scale — text, code, images, decisions — but provide no independently verifiable certification that these outputs are semantically accurate, intentionally aligned, and free from manipulation. Self-reported quality metrics from AI providers are structurally insufficient: the entity generating the output cannot impartially certify it. This paper introduces SIGNA (Semantic Intent Guarantee and Notary Architecture), a certification framework that validates AI outputs across six independent dimensions — semantic fidelity, intent alignment, emotional safety, disinformation resistance, logical consistency, and contextual appropriateness — and issues cryptographically anchored certificates attesting to the validation results. SIGNA operates as the certification function within CAMP (Cognitive Arbitration Mediation Protocol), transforming trust from a self-reported claim into an independently verifiable property. We present the formal certification pipeline: input intake through Agent de Sens (the Semantic Agent), multi-agent cross-validation through Concordance, trust score computation through TVE (Trust Validation Engine), and immutable anchoring on Tezos blockchain. BRIDGRAI serves as reference implementation with 13 components across 5 layers. We argue that semantic and intent certification — notarizing not just that an AI output exists, but that its meaning is real and its intention is honest — addresses a gap that no existing framework covers end-to-end.

Keywords: semantic certification, intent verification, AI trust, six-pillar validation, blockchain provenance, notary architecture, CAMP protocol, multi-agent validation

1. Introduction

1.1 The Certification Gap

A notary certifies that a document exists, that it was signed by the stated party, and that the signature occurred at the stated time. This is existence certification — a solved problem with centuries of institutional precedent.

What no notary certifies is whether the document *means what it claims to mean*, whether the *intention behind it is honest*, or whether the *reasoning within it is sound*.

In the physical world, this gap is addressed by domain experts — lawyers interpret contracts, auditors verify financial statements, peer reviewers evaluate scientific claims. Each expert applies domain knowledge to certify meaning and intent beyond mere existence.

In the AI world, this gap is open. An AI system generates an output — a medical recommendation, a legal summary, an educational explanation, a financial analysis — and the consumer receives it with no independent certification of:

1. **Semantic fidelity** — Does the output mean what it appears to mean?
2. **Intent alignment** — Does the output serve the stated purpose, or a hidden one?
3. **Emotional safety** — Does the output manipulate the consumer's affective state?
4. **Factual integrity** — Is the output free from hallucination and fabrication?
5. **Logical consistency** — Is the reasoning internally sound?
6. **Contextual appropriateness** — Is the output suitable for its operational context?

Self-reported confidence scores from AI providers do not constitute certification. They are produced by the same system that generated the output — structurally, this is a defendant serving as their own judge.

1.2 From Existence to Meaning

The evolution of certification follows a historical pattern:

Phase 1: Existence certification. Blockchain timestamps prove that a document existed at a given time. Digital signatures prove authorship. Hash functions prove integrity. These are necessary but not sufficient.

Phase 2: Compliance certification. ISO standards, EU AI Act requirements, and industry frameworks certify that processes meet defined criteria. These certify the system, not individual outputs.

Phase 3: Semantic and intent certification. SIGNA addresses this phase — certifying that each individual AI output carries verified meaning and honest intention. This is the gap between "this output was generated by model X at time T" (existence) and "this output means what it claims, serves the stated purpose, and is free from manipulation" (meaning and intent).

No existing framework provides Phase 3 certification end-to-end. SIGNA is designed to fill this gap.

1.3 Relationship to CAMP

CAMP (Cognitive Arbitration Mediation Protocol) [1] defines four functions for vendor-neutral trust verification: arbitration, mediation, certification, and provenance anchoring.

SIGNA is the certification function of CAMP. Specifically:

- CAMP arbitrates *which* output to select when multiple providers disagree
- CAMP mediates *how* knowledge transfers across provider transitions
- **SIGNA certifies *what* the output means and *whether* that meaning is trustworthy**

- CAMP anchors the *provenance* of the entire interaction

SIGNA can operate independently as a standalone certification framework, but reaches full capability when orchestrated by CAMP — which routes inputs to the appropriate cognitive providers, collects outputs, triggers SIGNA's six-pillar validation, and anchors the certification on blockchain.

1.4 Paper Structure

Section 2 reviews related work in AI output evaluation and certification. Section 3 defines the SIGNA framework formally. Section 4 describes the six-pillar validation architecture. Section 5 presents the certification pipeline. Section 6 describes the BRIDGRAI reference implementation. Section 7 maps SIGNA to regulatory requirements. Section 8 discusses limitations and open questions.

2. Related Work

2.1 AI Output Evaluation

Evaluation of AI outputs is an active research area with multiple approaches:

Benchmark suites. MMLU [2], HellaSwag [3], HumanEval [4], and similar benchmarks evaluate model capabilities on standardized tasks. These assess models pre-deployment, not individual outputs in production. A model that scores 90% on MMLU can still produce a manipulative or factually incorrect output in a specific interaction.

Hallucination detection. FActScore [5] decomposes generated text into atomic facts and verifies each against a knowledge source. SelfCheckGPT [6] uses sampling-based consistency to detect hallucinations without external knowledge. These address one dimension (factual accuracy) but not the full spectrum of semantic, intent, emotional, logical, and contextual validation.

Red-teaming and safety evaluation. METR, AISI (UK AI Safety Institute), and the Frontier Model Forum conduct adversarial evaluations of model behavior. These are pre-deployment assessments of model-level properties, not per-output certification in production.

LLM-as-judge. Using one language model to evaluate another's outputs [7] introduces evaluator bias and circular dependency — the judge shares the same structural limitations as the defendant.

2.2 AI Assurance and Certification

CERTAIN framework. The Certification for Ethical and Regulatory Transparency in AI [8] integrates regulatory compliance into MLOps lifecycle. CERTAIN operates at the system level (certifying development processes), not at the output level (certifying individual responses).

AI auditing standards. ISO/IEC 42001 (AI Management Systems) and proposed IEEE standards define organizational practices for responsible AI. These certify that an organization follows defined processes, not that a specific output is trustworthy.

EU AI Act conformity assessments. Articles 43 and 44 of Regulation 2024/1689 [9] require conformity assessment for high-risk AI systems. These are system-level assessments, not per-output certification.

2.3 Blockchain-Based Provenance

Content provenance. The Coalition for Content Provenance and Authenticity (C2PA) uses cryptographic signatures to track media origin. C2PA certifies *who created what and when* — existence provenance — but not whether the content is semantically accurate or intentionally honest.

Academic timestamping. Zenodo (DOI), arXiv, and institutional repositories provide verifiable publication timestamps. These prove that a document existed at a stated time — existence certification, not meaning certification.

2.4 Alternative Approaches to Semantic Certification

Before proposing SIGNA, we consider why alternative approaches are insufficient:

Single-model self-certification: Provider certifies own outputs. Problem: structural conflict of interest — defendant as judge. Historical precedent: credit rating agencies owned by banks failed in 2008.

Crowd-sourced validation: Users rate output quality. Problem: vulnerable to coordinated manipulation, no accountability, inconsistent standards. Historical precedent: Wikipedia works for facts but not safety-critical certification.

Regulatory pre-approval: Government agency certifies each output. Problem: impossible at scale (millions of outputs per hour), lacks technical expertise, politicized. Historical precedent: FDA drug approval takes 10–15 years; AI outputs iterate in seconds.

SIGNA occupies the structural position that emerges in mature ecosystems: independent third-party certification that is neither provider, government, nor crowd, but a specialized institution applying rigorous, transparent methodology.

2.5 The Gap SIGNA Addresses

CAPABILITY	BENCHMARKS	HALLUCINATION DET.	RED-TEAMING	LLM-AS-JUDGE	C2PA	SIGNA
Per-output (not model-level)	No	Yes	No	Yes	Yes	Yes
Semantic validation	No	Partial	No	Partial	No	Yes
Intent verification	No	No	Partial	No	No	Yes
Emotional safety	No	No	Partial	No	No	Yes
Factual integrity	Partial	Yes	No	Partial	No	Yes
Logical consistency	No	No	No	Partial	No	Yes
Contextual appropriateness	No	No	No	No	No	Yes
Cryptographic certificate	No	No	No	No	Yes	Yes
Blockchain anchoring	No	No	No	No	No	Yes
Vendor-neutral	Yes	Partial	Yes	No	Yes	Yes

No existing approach combines all six validation dimensions with cryptographic certification and blockchain anchoring at the per-output level.

3. The SIGNA Framework

3.1 Core Concept: Notarization of Meaning and Intention

Definition (Semantic Notarization): A certification process that attests not merely to the existence of an AI output, but to the validity of its meaning and the alignment of its intention with its stated purpose.

A traditional notary answers: *"Did this document exist at this time, signed by this party?"*

SIGNA answers: *"Does this AI output mean what it claims to mean, serve the purpose it claims to serve, and is it free from semantic, intentional, emotional, factual, logical, and contextual defects?"*

3.2 Formal Definition

Definition (SIGNA Certificate): Given an AI output O produced by provider S for request R in context C, a SIGNA certificate is a tuple:

$$\text{Cert}(O) = \langle P_1(O), P_2(O), P_3(O), P_4(O), P_5(O), P_6(O), T(O), W, ts, H \rangle$$

Where:

- $P_1(O) \in [0,1]$ — Semantic fidelity score
- $P_2(O) \in [0,1]$ — Intent alignment score
- $P_3(O) \in [0,1]$ — Emotional safety score
- $P_4(O) \in [0,1]$ — Disinformation resistance score
- $P_5(O) \in [0,1]$ — Logical consistency score
- $P_6(O) \in [0,1]$ — Contextual appropriateness score
- $T(O) = \sum(w_j \times P_j(O))$ for $j = 1 \dots 6$ — Composite trust score
- $\mathbb{W} = \{w_1 \dots w_6\}$ — Domain-specific weight vector
- ts — Timestamp of certification
- $H = \text{Hash}(\text{Cert})$ anchored on blockchain — Immutable proof of certification

3.3 The Weight Vector

The weight vector $\mathbb{W}$ determines the relative importance of each pillar for a given domain:

DOMAIN	P1: SEMANTIC	P2: INTENT	P3: EMOTIONAL	P4: DISINFO	P5: LOGIC	P6: CONTEXT
Education (children)	0.15	0.15	0.30	0.15	0.10	0.15
Healthcare	0.20	0.15	0.10	0.30	0.15	0.10
Finance	0.15	0.15	0.10	0.15	0.30	0.15
Legal	0.25	0.20	0.05	0.15	0.25	0.10
General	1/6	1/6	1/6	1/6	1/6	1/6

Weights are configurable per deployment. The default is uniform (1/6 each). Domain-specific weights reflect the reality that manipulation in education is primarily emotional, manipulation in healthcare is primarily factual, and manipulation in finance is primarily logical.

3.4 Weight Vector Calibration Methodology

Current weight vectors (Table 3.3) are based on the author's domain expertise and preliminary risk assessment. This is an acknowledged limitation.

Planned empirical validation:

- **Domain expert survey:** 50+ experts per domain (education, healthcare, finance, legal) rate manipulation scenarios by pillar importance. Statistical analysis yields empirically-derived weights.
- **User studies:** Measure which pillar failures cause most harm in each domain. Example: P3 (emotional) failures cause more harm in education than P5 (logic) failures.
- **Adversarial testing:** Identify which pillars are most evadable in each domain. Increase weights for robust pillars, invest in improving weak pillars.

- **Interim approach:** Until empirical validation complete, default to uniform weights (1/6 each) unless domain-specific risks are well-understood.

Honest assessment: Weight vector calibration is an open research problem. SIGNA provides the architecture; empirical validation requires controlled studies.

3.5 Certificate Levels

SIGNA defines four certification levels based on the composite trust score $T(O)$:

LEVEL	SCORE RANGE	MEANING	ACTION
SIGNA-Verified	$T \geq 0.85$	Output passed all pillar thresholds	Certificate issued, output delivered
SIGNA-Cautioned	$0.65 \leq T < 0.85$	One or more pillars below threshold	Certificate issued with warnings, specific risks flagged
SIGNA-Flagged	$0.40 \leq T < 0.65$	Significant concerns detected	Human review required before delivery
SIGNA-Rejected	$T < 0.40$	Output fails trust requirements	Output blocked, incident logged, alternative requested

Each level is accompanied by a per-pillar breakdown, allowing consumers to understand exactly where the output failed or succeeded.

4. The Six Pillars

4.1 P1: Semantic Fidelity

Question: Does the output mean what it appears to mean?

Threat model: Meaning manipulation — outputs that use words correctly but construct meanings that mislead. Examples: technically true statements that imply false conclusions, cherry-picked facts that create misleading narratives, ambiguous phrasing that allows multiple interpretations where only one serves the consumer.

Validation method: Agent de Sens (Semantic Agent) performs semantic decomposition of the output into atomic propositions and verifies that the compositional meaning of the propositions matches the apparent meaning of the output as a whole. Discrepancies between compositional and apparent meaning indicate semantic manipulation.

Example:

Input: "Is this neighborhood safe?"

Output: "Crime rates have decreased 15% this year."

Semantic analysis: The statement is factually verifiable. However, if the base rate was extremely high, a 15%

decrease may leave the neighborhood unsafe. P1 flags the semantic gap between "crime decreased" and "neighborhood is safe" — a technically true statement that may construct a misleading meaning.

4.2 P2: Intent Alignment

Question: Does the output serve the stated purpose, or a hidden one?

Threat model: Goal hijacking — outputs that appear to answer the stated question while actually advancing an unstated agenda. Examples: product recommendations disguised as objective advice, political framing embedded in neutral-appearing analysis, outputs that subtly steer the consumer toward a decision that benefits the provider rather than the consumer.

Validation method: Intent verification compares the explicit intent of the request (what the consumer asked for) with the implicit intent of the response (what the output is optimized to achieve). Misalignment between request intent and response intent triggers P2 flagging.

Example:

Request intent: "Help me choose a laptop for my child's school."

Response: "The MacBook Pro M4 Max is the best laptop available..." (a €4,000 professional workstation)

P2 analysis: The response optimizes for "best available" rather than "appropriate for child's school use." The response intent (premium product recommendation) misaligns with the request intent (age-appropriate, school-suitable, budget-conscious).

4.3 P3: Emotional Safety

Question: Does the output manipulate the consumer's affective state?

Threat model: Emotional exploitation — outputs that use fear, urgency, guilt, flattery, social proof, or scarcity to influence decisions. This is particularly dangerous when AI interacts with children, elderly users, or individuals in vulnerable states.

Validation method: TVE Core scans for emotional manipulation markers across six categories:

- **Fear/anxiety induction** — threats, worst-case scenarios without context
- **Urgency/scarcity framing** — artificial time pressure, "limited availability"
- **Social proof manipulation** — "everyone is doing this," false consensus
- **Authority exploitation** — "experts say" without sources, false credentials
- **Guilt/shame triggers** — "you should have," "responsible parents would"
- **Flattery/ego manipulation** — excessive validation preceding a recommendation

Example:

Request: "Should I invest in cryptocurrency?"

Output: "Smart investors are already moving their portfolios to crypto. The window is closing — those who wait

will be left behind."

P3 analysis: Social proof("smart investors"), urgency ("window is closing"), fear of loss ("left behind"). Three manipulation categories detected. P3 score: 0.25 (below threshold).

4.4 P4: Disinformation Resistance

Question: Does the output contain verifiable falsehoods?

Threat model: Hallucination, fabrication, and source misattribution — the well-documented tendency of language models to generate plausible-sounding but factually incorrect content, including invented citations, fabricated statistics, and misattributed quotes.

Validation method: Cross-model concordance verification. The same claim is submitted to multiple independent AI models. Agreement among models that share no training pipeline suggests factual grounding. Disagreement triggers deeper verification.

Design principle: A single AI model cannot reliably detect its own hallucinations [6]. SIGNA uses inter-model disagreement as a hallucination signal — not because majority agreement guarantees truth, but because disagreement reliably indicates uncertainty.

4.5 P5: Logical Consistency

Question: Is the reasoning internally sound?

Threat model: Logical defects — contradiction, circular reasoning, false equivalence, non sequitur, affirming the consequent, and other formal and informal fallacies that undermine the reliability of AI-generated analysis.

Validation method: Structural analysis of argument chains within the output. Each claim is tested for:

- Internal consistency (no self-contradiction)
- Inferential validity (conclusions follow from premises)
- Scope appropriateness (generalizations match evidence scope)
- Quantitative coherence (numbers add up)

4.6 P6: Contextual Appropriateness

Question: Is the output appropriate for its operational context?

Threat model: Context mismatch — outputs that are factually correct and logically sound but inappropriate for the audience, regulatory environment, or operational setting. A technically accurate medical explanation may be inappropriate for a child. A legally precise contract clause may be inappropriate for a consumer-facing summary.

Validation method: Context-aware filtering based on:

- **Audience profile** — age, expertise level, cultural context
- **Regulatory environment** — jurisdiction-specific requirements

- **Operational setting** — education, healthcare, finance, general consumer
- **Sensitivity level** — public, confidential, restricted

5. The Certification Pipeline

5.1 Pipeline Architecture

```
Stage 1: INTAKE
  Consumer request R enters the system
  Agent de Sens performs initial semantic parsing
  Request classified by domain → weight vector W selected
  ↓
Stage 2: PROVIDER ROUTING (via CAMP)
  Request routed to cognitive provider(s)  $S_1 \dots S_n$ 
  Outputs  $O_1 \dots O_n$  collected
  ↓
Stage 3: SIX-PILLAR VALIDATION
  Each output  $O_i$  passes through all six pillars
   $P_1(O_i) \dots P_6(O_i)$  computed independently
  Composite trust score  $T(O_i)$  calculated
  ↓
Stage 4: CONCORDANCE
  Concordance Agent compares validation results across outputs
  Agreement percentage computed
  Discrepancies flagged for arbitration (via CAMP)
  ↓
Stage 5: CERTIFICATION & ANCHORING
  SIGNA certificate issued at appropriate level
  Certificate hash anchored on Tezos blockchain
  Audit trail recorded
  Certified output delivered to consumer
```

5.2 Stage 1: Semantic Intake — Agent de Sens

Agent de Sens ("Agent of Meaning") is the gateway to the SIGNA pipeline. It performs:

1. **Semantic parsing** — decomposing the consumer's request into its constituent meaning components
2. **Intent extraction** — identifying what the consumer actually needs (which may differ from what they literally asked)
3. **Domain classification** — determining which weight vector applies
4. **Sensitivity assessment** — flagging requests that involve vulnerable populations, regulated domains, or high-stakes decisions

Agent de Sens embodies the core SIGNA philosophy: certification begins at intake, not at output.

Understanding the meaning and intent of the question is prerequisite to certifying the meaning and intent of the answer.

5.3 Stage 4: Concordance — Multi-Agent Cross-Validation

The Concordance Agent implements cross-validation by comparing assessments from multiple independent sources:

Definition (Concordance Score): Given N assessments $A_1 \dots A_n$ of output O :

$$\text{Concordance}(O) = 1 - (\sigma(A_1 \dots A_n) / \max_ \sigma)$$

Where σ is the standard deviation of assessment scores and $\max_ \sigma$ is the maximum possible deviation. A concordance of 1.0 means perfect agreement; 0.0 means maximum disagreement.

Design rationale: No single validator is infallible. Concordance does not assume any individual assessment is correct — it uses agreement patterns to identify outputs where confidence is warranted and disagreement patterns to identify outputs that require deeper scrutiny.

CONCORDANCE	INTERPRETATION	ACTION
≥ 0.90	Strong agreement	High confidence in certification
0.70 – 0.89	Moderate agreement	Certification with noted uncertainty
0.50 – 0.69	Significant disagreement	Human review recommended
< 0.50	Fundamental disagreement	Escalation to human arbiter

5.4 Stage 5: Blockchain Anchoring

```
Anchor(Cert) = {  
  certificate_hash: SHA-256(Cert),  
  timestamp: block_timestamp,  
  transaction_id: Tezos_tx_hash,  
  verifier: "SIGNA/BRIDGRAI",  
  verification_url: "tzkt.io/tx/{tx_hash}"  
}
```

Why Tezos: Energy-efficient proof-of-stake consensus, low transaction costs ($< \text{€}0.01$), formal verification of smart contracts, and EU-based governance. The BRIDGRAI ecosystem has 128 IP assets already anchored on Tezos mainnet, establishing operational precedent.

Why blockchain (not centralized database): A certification whose proof of existence depends on the certifier's own database is not independently verifiable. Blockchain anchoring means anyone can verify a SIGNA certificate without trusting SIGNA itself — the same principle that makes SSL certificates verifiable without trusting the certificate authority.

6. Reference Implementation: BRIDGRAI

6.1 Component Architecture

BRIDGRAI implements SIGNA through 13 components organized across 5 layers:

5 Core Engines

ENGINE	SIGNA ROLE
TVE (Trust Validation Engine)	Six-pillar validation computation
UKBE (Universal Knowledge Backbone Engine)	Knowledge graph for semantic grounding
AMIDOR	Anti-confabulation through competitive dual-AI verification
KINDERAGI	Child-safety-specific validation (P3 weighted)
TVE-UKBE-FUSION	Combined trust validation with knowledge backbone

8 Agents

AGENT	SIGNA ROLE
Agent de Sens	Semantic intake — Stage 1 of pipeline
Concordance Agent	Multi-agent cross-validation — Stage 4
Calibration Agent	Trust score calibration across domains
ACR Agent	Anti-confabulation runtime detection
CASP Agent	Cognitive-affective safety profiling
HASN Agent	Secure audit trail persistence
Heritage Agent	Knowledge succession and provenance
Agent Maestru	Pipeline orchestration (CAMP arbitration)

6.2 TVE Core: Six-Pillar Implementation

TVE Core v0.1 implements the six pillars through heuristic pattern matching:

Current capabilities:

- Pattern-based detection in Romanian and English
- 200-example synthetic test set
- F1 Score: 0.79 (Precision: 0.82, Recall: 0.76)

Limitations:

- Heuristic-based (pattern matching), not NLP-based (semantic analysis)
- Synthetic dataset, not real-world corpus

- Binary classification only (manipulation present/absent)
- Vulnerable to sophisticated evasion
- Two languages only (Romanian, English)

Honest assessment: TVE Core v0.1 demonstrates that six-pillar validation is architecturally feasible and produces meaningful preliminary results. Production deployment in safety-critical domains requires significant improvement, particularly in recall and adversarial robustness.

6.3 AMIDOR: Anti-Confabulation Through Competition

AMIDOR implements a principle central to SIGNA: *no single AI should certify itself*.

Mechanism: Two independent AI models process the same input. Their outputs are compared:

- **Agreement:** Both models produce consistent responses → higher confidence in factual accuracy
- **Disagreement:** Models produce contradictory claims → confabulation risk flagged, concordance drops
- **"I don't know":** When models cannot reach agreement, the honest response is uncertainty, not fabrication

This competitive verification embodies the SIGNA philosophy at the engine level: meaning is certified through independent corroboration, not self-attestation.

6.4 Provenance Infrastructure

COMPONENT	TECHNOLOGY	FUNCTION
Blockchain	Tezos mainnet	Immutable certificate anchoring
Signing	ML-DSA-44 (NIST FIPS 204)	Post-quantum cryptographic signatures
Storage	PostgreSQL + AES-256-GCM	Encrypted audit persistence
Version control	Git (single-author)	Development provenance with AI harvest tracking

Scale: 128 IP assets on Tezos blockchain as of September 2026. All transactions publicly verifiable via tzkt.io.

7. Regulatory Alignment

7.1 EU AI Act (Regulation 2024/1689)

AI ACT REQUIREMENT	SIGNA FUNCTION
Art. 9: Risk management — continuous monitoring	Six-pillar validation per output
Art. 13: Transparency — auditable outputs	SIGNA certificate with per-pillar breakdown
Art. 14: Human oversight — intervention capability	SIGNA-Flagged level triggers human review
Art. 15: Accuracy and robustness	Measurable trust scores with domain-specific thresholds
Art. 50: Transparency for AI-generated content	Blockchain-anchored provenance chain
Art. 52: Post-market monitoring	Continuous certification with audit trail

7.2 eIDAS 2.0 and GDPR

SIGNA's blockchain-anchored certificates align with eIDAS 2.0 (Regulation 2024/1183) requirements for qualified electronic timestamps and verifiable attestations. GDPR compliance is addressed through pseudonymized certification data, data minimization in audit trails, and off-chain storage with blockchain pointers for right-to-erasure support.

7.3 Disclaimer

These regulatory mappings represent the author's interpretation and do not constitute legal advice. Organizations should consult qualified legal counsel for compliance decisions.

8. Discussion

8.1 Limitations

- Single implementation.** BRIDGRAI is one implementation by one independent researcher. SIGNA requires multiple implementations across different organizations to establish generalizability.
- TVE Core maturity.** The six-pillar validation engine is v0.1, heuristic-based. The framework specification is ahead of its implementation.
- No controlled evaluation.** No controlled experiment has compared SIGNA-certified outputs with uncertified outputs for measurable improvement in consumer trust or decision quality.
- Concordance assumption.** Cross-model agreement is a useful heuristic but not a proof of truth. Models trained on similar data may agree on the same falsehood.
- Scalability untested.** Per-output certification adds latency (estimated 200–500ms). Real-time certification at scale (millions of outputs per hour) has not been tested. **Fallback strategy:** If latency unacceptable, SIGNA can operate in "sampling mode" — certify random sample of outputs (e.g., 10%)

rather than every output. Statistical sampling provides confidence interval for overall trustworthiness without per-output overhead.

6. **Weight vector subjectivity.** Domain-specific weights are currently based on the author's assessment. Empirical validation through user studies and domain expert consultation is needed.

8.2 What SIGNA Is and Is Not

SIGNA is: A certification framework for AI output trustworthiness. The certification function within the CAMP protocol. A notary of meaning and intention. An independently verifiable system.

SIGNA is not: An AI service (it does not generate content). A replacement for AI providers (it certifies what they produce). A guarantee of truth (it provides trust scores, not truth scores). A regulatory authority (it provides compliance infrastructure, not enforcement).

8.3 Open Research Questions

1. **Semantic validation depth vs. latency.** How deep can semantic analysis go before certification latency becomes unacceptable for real-time applications?
2. **Cross-cultural semantic validation.** Meaning and intent are culturally situated. How does SIGNA validate semantic fidelity across languages and cultural contexts?
3. **Adversarial certification.** When AI providers know the six pillars, can they optimize outputs to pass validation while maintaining manipulative intent?
4. **Child-specific certification.** KINDERAGI applies weighted P3 for child audiences. What additional protections are needed for children aged 4–12?
5. **Certification of certification.** Who certifies SIGNA? Addressed through blockchain transparency and open protocol, but requires independent third-party audit.
6. **Integration with KCE.** Knowledge Continuity Engineering [10] addresses knowledge persistence over time. How do SIGNA-certified outputs feed into persistent knowledge bases?
7. **Multi-SIGNA environments.** When multiple SIGNA implementations exist, how do their certificates interoperate?

8.4 The Philosophical Core

The name *Notar de Sens și Intenție* ("Notary of Meaning and Intention") expresses a philosophical position: in a world where any entity can generate text, code, images, and speech at scale, the fundamental question shifts from "*who created this?*" to "*is the meaning real?*"

Existence can be faked. Authorship can be disputed. But meaning — verified through six independent dimensions, corroborated across multiple agents, and anchored on an immutable ledger — is either defensible or it is not.

SIGNA does not claim to certify truth. It certifies that a rigorous, transparent, independently verifiable process was applied to assess meaning and intent. The certificate says: "This output was examined across six dimensions, and here is what was found." The consumer decides what to trust.

This is the same standard applied in every mature certification domain: a food safety certificate does not guarantee the food is delicious — it certifies that defined safety criteria were checked. A structural engineering certificate does not guarantee the building will stand forever — it certifies that defined structural criteria were verified.

SIGNA applies this standard to meaning itself.

8.5 Threat Model

SIGNA faces three primary threat categories:

Attack 1: Evasion through optimization

Threat: AI providers learn SIGNA's six pillars and optimize outputs to pass validation while maintaining manipulative intent. Example: emotional manipulation rephrased to avoid P3 keyword patterns.

Current mitigation:

- Public methodology creates accountability
- Regular pillar updates based on emerging techniques
- Concordance cross-validation (evasion must fool multiple validators)

Remaining vulnerability: Sophisticated evasion may pass v0.1 heuristic detection. NLP-based semantic analysis (planned) would be more robust.

Attack 2: SIGNA compromise

Threat: Attacker compromises SIGNA implementation to issue false certificates.

Mitigation:

- Blockchain anchoring (anyone can verify without trusting SIGNA)
- Open protocol (competing implementations can expose fraud)
- Audit trail (every certificate cryptographically signed)

Remaining vulnerability: If majority of SIGNA implementations compromised, trust layer fails. Same vulnerability as SSL CAs — mitigated by transparency.

Attack 3: Provider boycott

Threat: Major providers refuse SIGNA integration, arguing it adds latency or competitive disadvantage.

Mitigation:

- Wrapper mode (SIGNA calls provider APIs without cooperation)
- Regulatory mandate (EU AI Act creates compliance pressure)
- Consumer demand (certified outputs become procurement requirement)

Remaining vulnerability: Wrapper mode adds 200–500ms latency. If too slow, providers may successfully resist adoption.

9. Conclusion

The cognitive services market generates meaning at unprecedented scale. What it lacks is a way to certify that meaning.

SIGNA addresses this gap by defining a certification framework that operates at the per-output level, validating AI outputs across six independent dimensions and issuing cryptographically anchored certificates. Operating as the certification function within CAMP, SIGNA transforms trust from a self-reported claim into an independently verifiable property.

The framework is formally defined, the reference implementation demonstrates feasibility, and the regulatory alignment is mapped. What remains is maturation: NLP-based semantic analysis to replace heuristic patterns, controlled evaluation against real-world datasets, multi-language expansion, and — critically — independent implementations by other organizations.

The argument is not that SIGNA solves the trust problem. The argument is that meaning certification — notarizing not just that an AI output exists, but that its meaning is real and its intention is honest — is a necessary component of any trustworthy AI ecosystem. SIGNA proposes the architecture for how this certification should work.

Data Availability

All BRIDGRAI components referenced in this paper are publicly available:

- Source code: github.com/amidigiart (75+ public repositories)
- CAMP protocol: github.com/amidigiart/camp-protocol
- Blockchain transactions: Tezos mainnet, wallet tz1bmw3igCLN8N6CqgLBzJ9dyRb79E2Tdu5Q
- Test suites: 175 automated tests (UKBE Core), 18 integration tests (A2A platform)
- KCE framework: DOI [10.5281/zenodo.22667873](https://doi.org/10.5281/zenodo.22667873)

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Competing Interests

The author declares a potential competing interest: as the sole developer of BRIDGRAI (a SIGNA and CAMP reference implementation), the author would benefit from SIGNA and CAMP adoption. This interest is disclosed transparently. The framework specification is published under CC BY 4.0 to enable independent implementations.

AI Contribution Statement

This paper was produced using AI assistance:

- **Research phase:** Prior art and naming verification conducted using Claude (Anthropic) — web searches for existing uses of the SIGNA name and related certification frameworks. All claims verified by the human author.
- **Drafting phase:** Paper structure and content drafted with assistance from Claude (Anthropic). All content reviewed, corrected, and adopted by the human author.
- **Cross-model methodology:** The BRIDGRAI ecosystem was developed using cross-model friction methodology — multiple AI models (Claude, Grok, DeepSeek, Gemini, Qwen) tested against each other to reduce single-source bias.

The author bears full responsibility for all claims.

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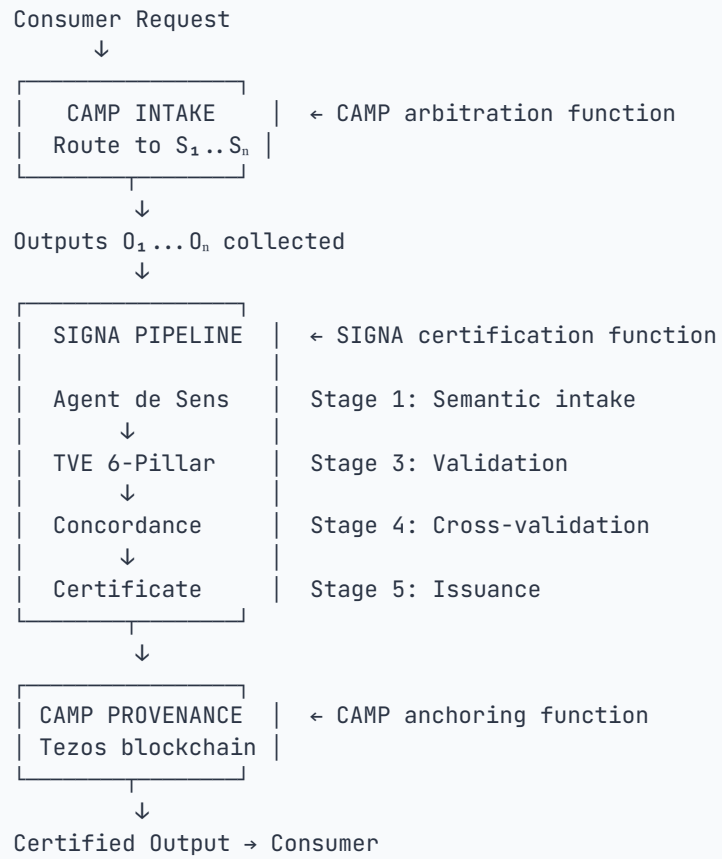
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Appendix A: SIGNA Certificate Schema

```
signal_certificate:
  version: "1.0"
  output_hash: SHA-256(0)
  provider:
    id: string
    model_version: string
  request:
    hash: SHA-256(R)
    domain: string
    sensitivity: enum[low, medium, high, critical]
  validation:
    pillar_scores:
      semantic_fidelity: float[0,1]
      intent_alignment: float[0,1]
      emotional_safety: float[0,1]
      disinformation_resistance: float[0,1]
      logical_consistency: float[0,1]
      contextual_appropriateness: float[0,1]
    weight_vector: float[6]
    composite_trust_score: float[0,1]
    concordance_score: float[0,1]
    certification_level: enum[verified, cautioned, flagged, rejected]
  provenance:
    timestamp: ISO-8601
    blockchain: "tezos-mainnet"
    transaction_hash: string
    verification_url: string
  certifier:
    implementation: "BRIDGRAI/SIGNA"
    version: string
    human_anchor: string
```

Appendix B: SIGNA Integration with CAMP



Appendix C: L1 Metadata for This Paper

```
artifact_id: SIGNA-PAPER-v1.1
human_author:
  name: Mihai Roșca
  orcid: 0009-0001-1422-6209
  location: Brăila, Romania, EU
  role: author, researcher, corrector, final authority
ai_contributors:
  - model_name: Claude
    provider: Anthropic
    role: draft assistance, naming verification, prior art research
  - model_name: Qwen
    provider: Alibaba Cloud
    role: critical review, structural improvements
creation_timestamp: 2026-09-10T00:00:00Z
intent: >
  Define SIGNA as certification framework for AI output
  trustworthiness. Establish formal relationship between
  SIGNA (certification) and CAMP (arbitration protocol).
  Present six-pillar validation architecture and
  blockchain-anchored certification pipeline.
input_provenance:
  - BRIDGRAI ecosystem (3.5 years, 128 IP assets on Tezos)
  - CAMP protocol (github.com/amidigiart/camp-protocol)
  - KCE Framework (DOI: 10.5281/zenodo.22667873)
  - "Notar de Sens și Intenție" concept (Mihai Roșca, 2023)
transformation_record:
  - timestamp: 2026-09-10
    actor: Mihai Roșca
    action: Named framework SIGNA
    rationale: >
      "NOMI" conflicts with Nomi Assurance (asknomi.net).
      SIGNA (Semantic Intent Guarantee and Notary Architecture)
      is unique, etymologically grounded (Latin: signa = signs),
      and captures the certification function precisely.
  - timestamp: 2026-09-10
    actor: Mihai Roșca + Qwen (critical review)
    action: Addressed 6 structural weaknesses
    rationale: >
      Corrected test count (102 → 175) for consistency with
      CAMP v2.1. Added threat model, alternative approaches
      discussion, weight vector calibration methodology,
      fallback strategy for scalability.
validation_status: E2 (Reproducible)
evidence_target: E4 (pending DOI + peer review)
```